

Macronutrient balance and lifespan

.

Dietary restriction (DR) without malnutrition is widely regarded to be a universal mechanism for prolonging lifespan. It is generally believed that the benefits of DR arise from eating fewer calories (termed caloric restriction, CR). Here we argue that, rather than calories, the key determinant of the relationship between diet and longevity is the balance of protein to non-protein energy ingested. This ratio affects not only lifespan, but also total energy intake, metabolism, immunity and the likelihood of developing obesity and associated metabolic disorders. Among various possible mechanisms linking macronutrient balance to lifespan, the nexus between the TOR and AMPK signaling pathways is emerging as a central coordinator.